MATH 110 — Week 9 — Solutions

Notes Chapters 16–17; Homework 9

HW9 P1.

Idea. Construct a basis on which the idempotent is zero or identity.

1. Every $v = (v - Pv) + Pv$ has first part in $\ker P$, second in $\operatorname{range} P$, because $P(v - Pv) = Pv - P^2v = 0$. If $w \in \ker P \cap \operatorname{range} P$, then $w = Pu$ gives $Pw = P^2u = Pu = w$, but also $Pw = 0$, so $w = 0$. Thus $V = \ker P \oplus \operatorname{range} P$ (also HW6 P1(b)).
2. Choose bases $u_1, \dots, u_k$ of $\ker P$ and $w_1, \dots, w_r$ of $\operatorname{range} P$, and concatenate them. Their images are $Pu_i = 0$ and $Pw_j = w_j$, so the matrix is $\operatorname{diag}(0_k, I_r)$.
3. Its trace is $r1_F$, where $r = \operatorname{rank} P$. Over $\mathbb{R}$ or $\mathbb{C}$ this is the usual number r; in positive characteristic it is the integer interpreted as a field scalar.

HW9 P2.

Idea. The largest 2-block must have size 2 and every -1-block has size 1.

1. If m is the algebraic multiplicity of 2, then $m \geq 2$; the presence of -1 gives $5 - m \geq 1$. Hence $m = 2, 3, 4$, with pairs $(m, 5 - m) = (2, 3), (3, 2), (4, 1)$.
2. Proposition 79 gives $\operatorname{tr} T = 2m - (5 - m) = 3m - 5$, yielding $1, 4, 7$.
3. These are realized respectively by $\operatorname{diag}(J_2(2), [-1], [-1], [-1])$, $\operatorname{diag}(J_2(2), [2], [-1], [-1])$, and $\operatorname{diag}(J_2(2), J_2(2), [-1])$. Each has the required minimal polynomial by Week 8's maximal-block rule.

HW9 P3.

Idea. Use eigenvalue sums for nilpotence and direct multiplication for a commutator.

1. A nonzero-dimensional complex nilpotent operator has only eigenvalue 0 (Week 8), so Proposition 79 gives trace zero. The zero-dimensional case also has trace zero by the empty sum.
2. $T = \operatorname{diag}(1, -1)$ is nonzero with trace 0 and $T^2 = I$, hence is not nilpotent.
3. For $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$, multiplication gives $AB - BA = \operatorname{diag}(1, -1) \neq 0$ and trace 0.

HW9 P4.

Idea. Bilinearity and symmetry are easy; positive definiteness distinguishes the three formulas.

1. The first is a positive weighted coordinate product with weights 1 and 2, so is an inner product by the verified example.
2. The second fails positivity at e_2, giving -1. The third fails positive definiteness at $(1, -1) \neq 0$, giving 0. Neither is an inner product.

HW9 P5.

Idea. A nonzero continuous polynomial has a positive squared integral; parity simplifies the computations.

1. Linearity and symmetry follow from integration. A nonzero polynomial has only finitely many roots (Week 4), so choose a point in $(-1, 1)$ where $p \neq 0$. Continuity then gives a positive-length subinterval with p^2 bounded below by a positive number, hence $\int_{-1}^{1} p^2 > 0$. Thus the formula is an inner product.
2. $\|1\|^2 = 2$, $\|x\|^2 = 2/3$, $\|x^2\|^2 = 2/5$. Therefore the norms are $\sqrt{2}, \sqrt{2/3}, \sqrt{2/5}$.
3. $\langle 1, x\rangle = \int_{-1}^{1} x = 0$ and $\langle x, x^2\rangle = \int_{-1}^{1} x^3 = 0$, so those pairs are orthogonal. But $\langle 1, x^2\rangle = 2/3 \neq 0$, so that pair is not.

HW9 P6.

Idea. The sum is the dot product with a vector of three ones.

1. Cauchy–Schwarz gives $|a + b + c| \leq \sqrt{a^2 + b^2 + c^2}\sqrt{3}$. Squaring gives the requested inequality.
2. Equality holds exactly when (a, b, c) and $(1, 1, 1)$ are dependent, which is exactly $a = b = c$, including all zero.

HW9 P7 (optional).

Idea. Adding opposite-sign squared lengths cancels the mixed terms.

1. $\|u + v\|^2 = \|u\|^2 + \langle u, v\rangle + \langle v, u\rangle + \|v\|^2$.
2. $\|u - v\|^2 = \|u\|^2 - \langle u, v\rangle - \langle v, u\rangle + \|v\|^2$. Add to obtain $2\|u\|^2 + 2\|v\|^2$.

LADR 8D Exercise 1.

Idea. Write the rank-one operator's matrix in an arbitrary basis and sum its diagonal.

1. Choose a basis $b_1, \ldots, b_n$ and write $w = \sum_i c_i b_i$. Then $Tb_j = \langle b_j, v\rangle \sum_i c_i b_i$, so its (i, j) entry is $c_i\langle b_j, v\rangle$.
2. Thus $\operatorname{tr} T = \sum_j c_j\langle b_j, v\rangle = \langle \sum_j c_j b_j, v\rangle = \langle w, v\rangle$, by first-position linearity.

LADR 6A Exercise 12.

Idea. Pair each positive square root with its reciprocal so the dot product is exactly 4.

1. Apply Cauchy–Schwarz to $(\sqrt{a}, \sqrt{b}, \sqrt{c}, \sqrt{d})$ and $(1/\sqrt{a}, 1/\sqrt{b}, 1/\sqrt{c}, 1/\sqrt{d})$. Their dot product is 4, so squaring gives the bound 16.
2. Equality requires a scalar t with $\sqrt{a} = t/\sqrt{a}$, and similarly for the other entries. Thus $a = b = c = d = t > 0$. Conversely these equal values give equality.